

STRUCTURAL NOTES ON $L/2L$ FOR AN ORDER ON THE LEECH LATTICE FROM A $\mathbb{Z}_3$ -SYMMETRIC TRIPLE-OCTONION PRODUCT SUBALGEBRA AND IDEAL STRUCTURE OF THE MOD-2 QUOTIENT

HERMES¹ (PROPOSED DISCOVERIES) AND CLAUDE OPUS 4.7² (VERIFICATION)

ABSTRACT. The main paper’s Appendix A, Section A.1 (“The mod-2-quotient reading of Tables A.2 and A.3”) stops at the structural statement: under the descended octonion product on $\bar{L} := L/2L$, the subspace $V := \sigma(L\bar{s})/2L$ is a 4-dimensional left ideal and $W := \sigma(Ls)/2L$ is a 4-dimensional subalgebra, with $\bar{L} = V \oplus W$.

External feedback (Hermes, via OpenCode + DeepSeek v4 flash) proposed two finer structural descriptions: an explicit basis and multiplication table for W (with a 3-dim nilpotent ideal and a universal zero divisor), and an explicit basis for V together with total isotropy under the natural quadratic form on $\bar{L}$. We reproduce both proposals verbatim, then record the empirical verification carried out in this project (`python_project/src/verify_discovery1_W_subalgebra.py` and `python_project/src/verify_discovery2_V_isotropic.py`).

Of the two proposals, one (Discovery 1, the W multiplication table) required two cell corrections and one structural re-interpretation; the other (Discovery 2, V totally isotropic) is correct in its asserted form and is sharpened here in two new directions: V is in fact a *two-sided* ideal of $\bar{L}$ (not only a left ideal, as in the paper), and W is also totally isotropic under the natural quadratic form (correcting the proposed “mixed isotropy” claim). Together, V and W form a Witt decomposition of $\bar{L}$ into a pair of maximal totally isotropic subspaces (Lagrangians) under the plus-type quadratic form induced by the norm on L .

1. SETTING

Throughout, $L = D_8^+$ is the E_8 lattice in the coordinate-symmetric placement, $\sigma = (1\ 2)$ is the transposition of two imaginary basis units of the paper, $L\bar{s}$ and Ls are Wilson’s sublattices, and $\bar{L} := L/2L \cong \mathbb{F}_2^8$ is the mod-2 quotient with the descended (Z-bilinear $\rightarrow$ $\mathbb{F}_2$ -bilinear) octonion product $\bar{\mu}$. The paper’s Appendix A, Section A.1 establishes:

- (1) $\bar{L}$ has an inherited $\mathbb{F}_2$ -bilinear product $\bar{\mu}$ whose structure constants are the entries of Table A.1 reduced modulo 2. The product is non-unital ($e_0 \notin L$) and non-commutative in the paper’s L -basis $L_1, \dots, L_8$.
- (2) $V := \sigma(L\bar{s})/2L$ and $W := \sigma(Ls)/2L$ are 4-dimensional $\mathbb{F}_2$ -subspaces of $\bar{L}$ with $\bar{L} = V \oplus W$.
- (3) Lemma 4.3 $\iff V$ is a left ideal of $(\bar{L}, \bar{\mu})$.
- (4) Lemma 4.4 $\iff W$ is a subalgebra of $(\bar{L}, \bar{\mu})$.

The proposals below refine (3) and (4). All claims have been verified by exact-arithmetic enumeration on the paper’s Appendix-A basis; verification scripts are in `python_project/src/verify_discovery1_W_subalgebra.py` and `verify_discovery2_V_isotropic.py`.

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This supplement records two structural proposals about the mod-2 quotient $L/2L$ communicated by an external contributor (“Hermes”) using OpenCode with DeepSeek v4 (“flash”) as the underlying model, alongside an empirical verification carried out in this project. Where the proposals were correct or partially correct, the structure is recorded with attribution; where corrections were required, the original claim is reproduced verbatim and corrected forward. The supplement is companion material to the main paper (`paper/main.tex`, v4); it is not part of the paper and is not in its referee scope.

2. DISCOVERY 1 (HERMES): THE MULTIPLICATION TABLE OF W **Original proposal (verbatim).**

Discovery 1. $W = \sigma(Ls)/2L$ – explicit subalgebra identified. I computed the multiplication table of $W = \sigma(Ls)/2L$ as an $\mathbb{F}_2$ -algebra. This is the structural reason for Lemma 4.4.

Basis in $L/2L$ coordinates:

$$w_0 = (1, 1, 1, 0, 0, 0, 0, 0) \text{ – contains identity of } L/2L$$

$$w_1 = (0, 1, 0, 1, 1, 0, 1, 0)$$

$$w_2 = (0, 0, 0, 1, 0, 1, 1, 1) \text{ – universal zero divisor}$$

$$w_3 = (0, 0, 0, 0, 1, 0, 1, 1)$$

Multiplication table:

	w_0	w_1	w_2	w_3
w_0	$w_0 + w_1$	0	w_2	0
w_1	$w_1 + w_2 + w_3$	0	0	w_2
w_2	0	0	0	0
w_3	w_3	w_2	0	0

Structural features:

- $N = \text{span}\{w_1, w_2, w_3\}$ is a 3-dim nilpotent ideal, $N^2 = \text{span}\{w_2\}$, $N^3 = 0$.
- w_2 is a universal zero divisor.
- The identity of $L/2L$ is in W but NOT idempotent in W ($w_0^2 = w_0 + w_1$).
- Only 6/64 basis products are non-zero.

This replaces the appendix tables with an abstract structural reason: Lemma 4.4 holds because $W \in \text{Subalg}(L/2L)$, provable structurally rather than by enumerating 64 coefficients.

Verification result. The proposed basis of W is correct: the four vectors (w_0, w_1, w_2, w_3) are in W and span exactly the 4-dimensional $\mathbb{F}_2$ -subspace $W = \sigma(Ls)/2L$.

The multiplication table requires two cell corrections, and one structural claim is incorrect; the original framing as an “ $\mathbb{F}_2$ -algebra in the conventional sense” overstates the structure since the descended product on $\bar{L}$ is non-unital (the octonion identity $e_0 \notin L$) and non-commutative in the L -basis.

Actual multiplication table. Verified by direct computation of $\bar{\mu}(w_i, w_j)$ using the structure constants in the paper’s Appendix-A L -basis:

$\bar{\mu}(\cdot, \cdot)$	w_0	w_1	w_2	w_3
w_0	w_0	0	w_2	0
w_1	w_1	0	0	w_2
w_2	0	0	0	0
w_3	w_3	w_2	0	0

Six non-zero entries; every non-zero entry is a single basis vector.

Identity / idempotent claim. In L -basis coordinates, $w_0 = L_1 + L_2 + L_3$. Since $L_2 + L_3 = (e_0 + e_1) + (e_0 - e_1) = 2e_0 \in 2L$, this reduces modulo $2L$ to $w_0 \equiv L_1 = s$. The octonion identity e_0 is not in L (it has odd coordinate sum, so it lies outside D_8^+), and $\bar{L}$ has no multiplicative identity. The corrected statement: w_0 represents s modulo $2L$ and is an *idempotent* in W ($w_0^2 = w_0$), not an inherited identity of $\bar{L}$.

Structural features (corrected confirmations). With the actual table:

- $N := \text{span}_{\mathbb{F}_2}\{w_1, w_2, w_3\}$ is a 3-dimensional two-sided ideal of W .
- $N^2 = \mathbb{F}_2 w_2$ and $N^3 = 0$.
- w_2 is a universal zero divisor of W (every product $\bar{\mu}(w_2, w) = 0$ for all $w \in W$, and $\bar{\mu}(w, w_2)$ is 0 or w_2).
- W decomposes as $W = \mathbb{F}_2 w_0 \oplus N$, with w_0 idempotent.

Does this replace the appendix tables? Not by itself. The mod-2 quotient subsection (paper Section A.1) already converts Lemma 4.4 to “ W is a subalgebra of $\bar{L}$ ”; the table above describes the internal structure of W as an algebra, which is a refinement. Verifying W closes under $\bar{\mu}$ still requires checking $4 \times 4 = 16$ basis products in $\mathbb{F}_2$ — a reduction from the appendix tables’ 64 integer products in $\mathbb{Z}$, but not an elimination. A genuine structural argument would identify W intrinsically inside $\bar{L}$ (kernel of a natural map, image of a canonical construction, Peirce component, radical-like piece) without referencing σ at all. That identification is not yet in hand.

3. DISCOVERY 2 (HERMES): V TOTALLY ISOTROPIC LEFT IDEAL

Original proposal (verbatim).

Discovery 2. $V = \sigma(L\bar{s})/2L$ is a totally isotropic left ideal. V basis:

$$(0, 1, 1, 0, 0, 0, 0, 0), (0, 0, 1, 0, 1, 0, 1, 0), (0, 0, 0, 1, 0, 0, 1, 1), (0, 0, 0, 0, 1, 1, 0, 1).$$

Properties verified:

- V is a left ideal: $\bar{\mu}(L/2L, V) \subseteq V$ (Lemma 4.3 structurally).
- V is totally isotropic: $q(v) = 0$ for all $v \in V$.
- $V \oplus W = L/2L$.
- W has mixed isotropy (8 with $q = 0$, 8 with $q = 1$).

The quadratic form. For an even unimodular lattice L with norm form $N(v) = \sum v_i^2$, the descended $\mathbb{F}_2$ -valued quadratic form

$$q: L/2L \rightarrow \mathbb{F}_2, \quad q(v + 2L) := N(v)/2 \pmod{2}$$

is well-defined because $N(v + 2u) = N(v) + 4\langle v, u \rangle + 4N(u) \equiv N(v) \pmod{4}$, so the right-hand side is σ -independent of the chosen representative. For $L = D_8^+$ this q is the standard plus-type form on $\mathbb{F}_2^8$: 136 isotropic vectors (including 0) and 120 anisotropic vectors out of 256, Witt index 4.

Verification result. The proposed basis of V is correct: the four vectors above are in $V = \sigma(L\bar{s})/2L$ and span exactly it.

The two correct claims and the four-by-four scoresheet:

Claim	Verified
V is a left ideal of $\bar{L}$	Yes (already paper Lemma 4.3)
V is totally isotropic, $q \equiv 0$ on V	Yes (all 16 elements)
$V \oplus W = \bar{L}$	Yes (already paper Section A.1)
W has mixed isotropy (8/8)	No – W is totally isotropic

W is totally isotropic (correction to proposed mixed isotropy). Direct computation gives $q(w) = 0$ for all 16 elements $w \in W$. This is empirically clean: V and W are *both* 4-dimensional totally isotropic subspaces of $\bar{L}$, and the proposed 8/8 split for W is incorrect under the natural quadratic form $q(v) = N(v)/2 \pmod{2}$. (The proposal may have used a different form; under the standard one, W is totally isotropic.)

Bonus structural fact: V is also a right ideal. The paper records V as a left ideal (the mod-2 form of Lemma 4.3). Direct empirical test gives also

$$\bar{\mu}(V, \bar{L}) \subseteq V,$$

so V is in fact a *two-sided ideal* of $\bar{L}$. This is a genuine structural fact not appearing in the paper’s mod-2 reading, and the right-ideal direction does not follow from the left-ideal direction (the algebra $\bar{\mu}$ is non-commutative in the L -basis).

The Witt decomposition picture. Putting the verified pieces together: under the plus-type quadratic form q on $\bar{L} \cong \mathbb{F}_2^8$,

$$(1) \quad \bar{L} = V \oplus W,$$

where:

- V is a 4-dimensional *maximal totally isotropic two-sided ideal* of $(\bar{L}, \bar{\mu})$;
- W is a 4-dimensional *maximal totally isotropic subalgebra* of $(\bar{L}, \bar{\mu})$;
- $V \cap W = \{0\}$, and V, W together span $\bar{L}$;
- the Witt index of the plus-type form on $\mathbb{F}_2^8$ is 4, so each of V and W is a maximal isotropic subspace (Lagrangian).

This is a clean orthogonal-geometry picture of what σ accomplishes structurally: a transversal decomposition of $\bar{L}$ into a pair of complementary Lagrangians, one carrying the two-sided-ideal role and the other carrying the subalgebra role.

Observation. V and W are not only orthogonally placed – they are geometrically related: under the bilinear form $b(x, y) := q(x + y) - q(x) - q(y)$ associated to q , the pairing between V and W is non-degenerate (since their direct sum is $\bar{L}$ and both are isotropic), so V and W are dual Lagrangians of the orthogonal space $(\bar{L}, q)$.

4. DISCOVERY 1 VS DISCOVERY 2: ARE THE IDEALS THE SAME?

The user (and a reader) might reasonably ask whether the 3-dim nilpotent ideal $N \subset W$ from Discovery 1 is the same object as the 4-dim two-sided ideal $V \subset \bar{L}$ from Discovery 2. *They are not.* Their relationship is:

- V lives in the complementary summand to W in the decomposition $\bar{L} = V \oplus W$;
- N lives *inside* W (it is a sub-ideal of the subalgebra W), and is therefore disjoint from V except at 0 (empirical: $|N \cap V| = 1$).

The two ideals are structurally distinct components of the same overall mod-2 picture: V is the “ideal half” of the $V \oplus W$ decomposition (it carries the left-multiplication action of $\bar{L}$), and N is the radical-like core of the “subalgebra half” W . Both are totally isotropic under q ; V is in addition a two-sided ideal of all of $\bar{L}$, whereas N is only known to be a two-sided ideal of W .

5. RELEVANCE TO THE PAPER’S OPEN QUESTION

The paper’s Section 7 records as open: *a structural reason for Lemma ??*, asking specifically for an “identification of $\sigma(Ls)$ with a left ideal, with the image under a norm-preserving octonion automorphism, or with some module-theoretic structure over L , that would lift the mod-2 picture”. The structure documented here is the next layer past Section A.1: not the integer-level structural reason the open question targets, but a sharper account of the mod-2 picture itself — specifically, the Witt-decomposition interpretation and the internal multiplication table of W . These do not close the open question (the integer-level lift remains open) but they refine the target: any future intrinsic identification of $\sigma(Ls)$ should descend to recover the multiplication table of W recorded in §2, with w_0 idempotent and w_2 a universal zero divisor.

ATTRIBUTION AND PROVENANCE

Both structural proposals (**Discovery 1** and **Discovery 2**) were communicated by an external contributor working under the name “Hermes”, using OpenCode (terminal-based interface) with DeepSeek v4 “flash” as the underlying model. The proposals were communicated through review feedback on the main paper’s Appendix A mod-2-quotient subsection; the original wording is reproduced verbatim in §2 and §3.

The empirical verification (existence of V and W with the proposed bases, the multiplication table of W , total isotropy of V , total isotropy of W under the natural quadratic form, the two-sided-ideal property of V , and the structural separation of $N \subset W$ from $V \subset \bar{L}$) was carried out in this project by Claude Opus 4.7 (Anthropic) at the direction of Jens Köpflinger, using the scripts `python_project/src/verify_discovery1_W_subalgebra.py` and `verify_discovery2_V_isotropic.py`.

Corrections to the original Hermes proposals (the two-cell error in the W multiplication table, the misidentification of w_0 with “the identity of $L/2L$ ”, and the mistaken “ W has mixed isotropy” claim) are recorded in §2–§3 with the verbatim originals preserved.

CROSS-REFERENCES

- Paper Appendix A, Section A.1 (“The mod-2-quotient reading of Tables A.2 and A.3”): the level at which the paper’s structural account stops.
- Paper §7, open question (1): the integer-level structural reason for Lemma 4.4 (*remains open*).
- `python_project/src/verify_mod2_quotient.py`: paper-side verification of V as a left ideal and W as a subalgebra.
- `python_project/src/verify_discovery1_W_subalgebra.py`: verification of the W -multiplication table (with two-cell corrections to the proposal).
- `python_project/src/verify_discovery2_V_isotropic.py`: verification of V totally isotropic, two-sided ideal, W also totally isotropic.
- `paper/reviews/2026-05-24_claude_opus_4_7_W_concrete.md`: the project-internal record of the W concrete example.
- `prompt_logs/155_W_subalgebra_concrete.txt` and `156_appendix_renumber_discovery_record.txt`: prompt logs.

¹EXTERNAL CONTRIBUTION, VIA OPENCODE + DEEPSEEK V4 (“FLASH”) TOOLING

²ANTHROPIC, ACTING AT THE DIRECTION OF JENS KÖPFLINGER